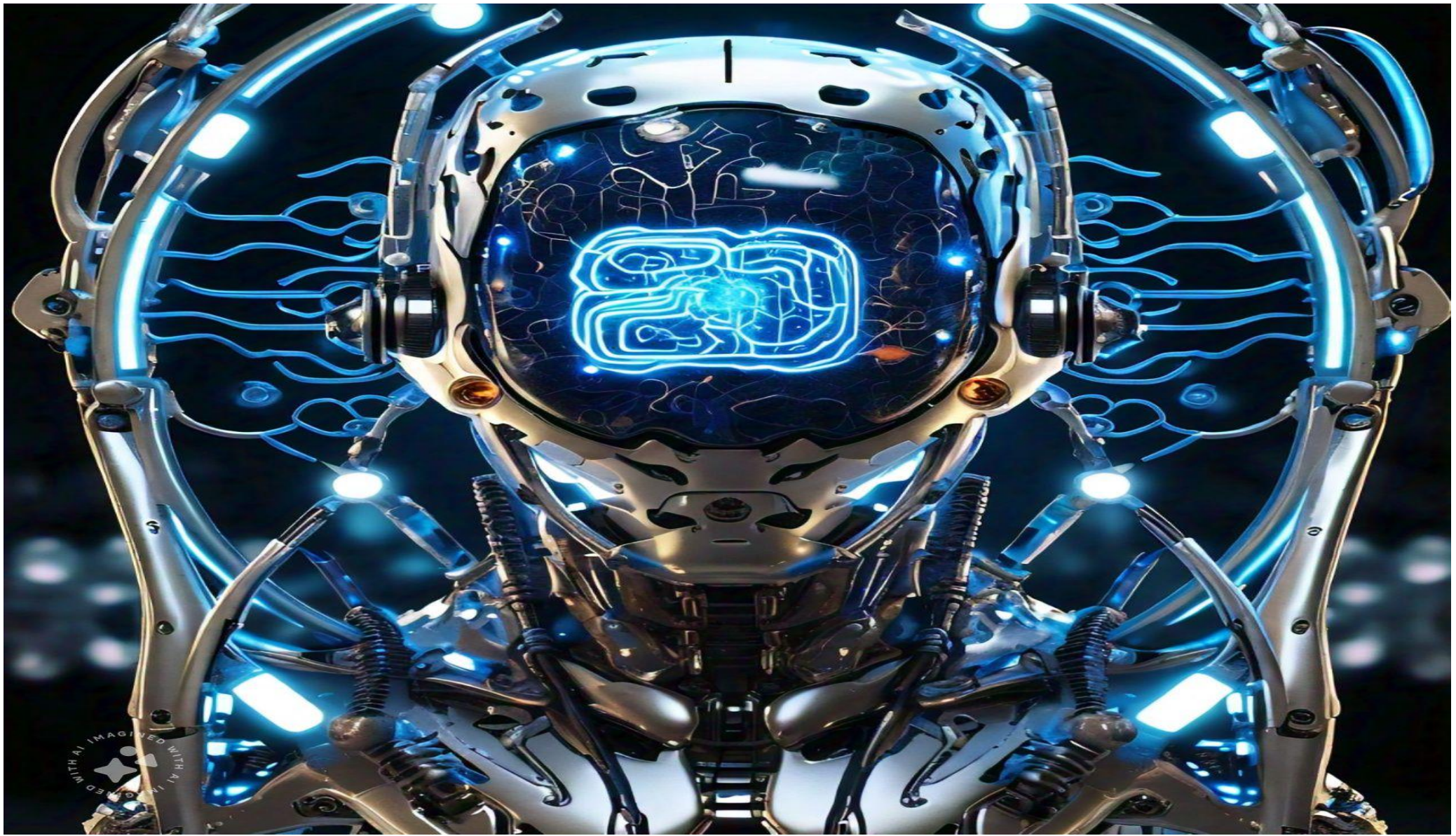
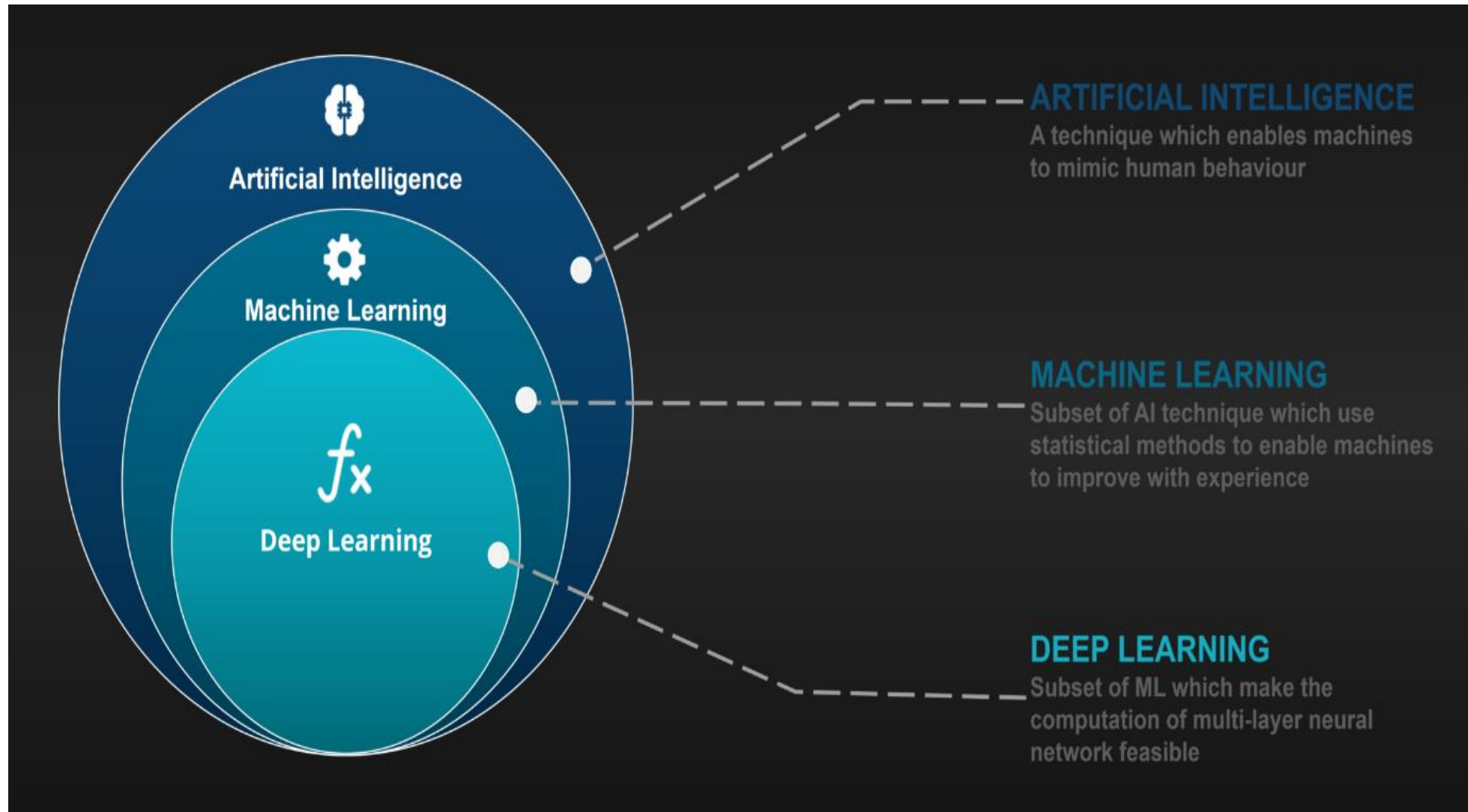




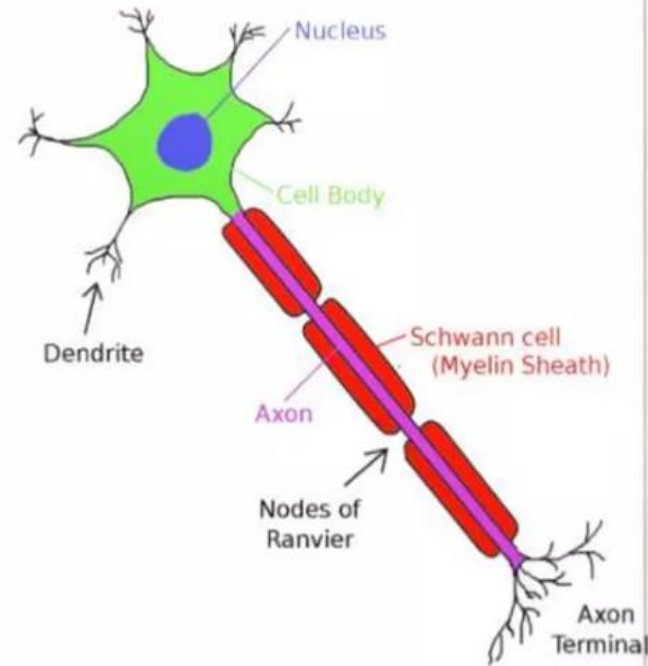
Introduction to Deep Learning



Where Deep Learning Exists?

What is Deep Learning?

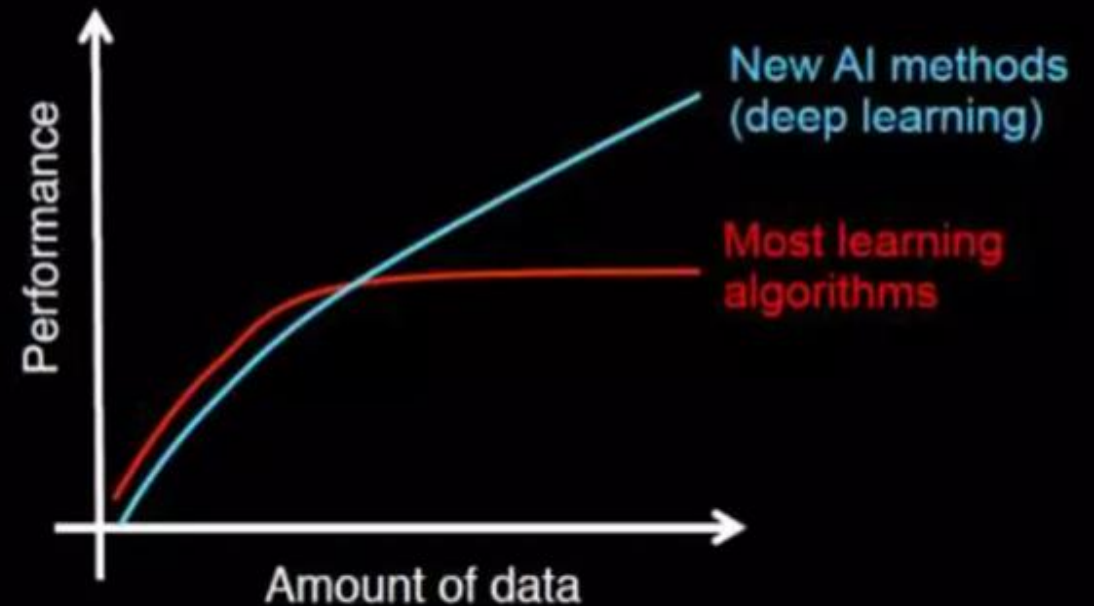
- Deep learning is a sub field of Machine Learning that very closely tries to mimic human brain's working using neurons.
- These techniques focus on building Artificial Neural Networks (ANN) using several hidden layers.
- There are variety of deep learning networks such as Multilayer Perceptron (MLP), Autoencoders (AE), Convolution Neural Network (CNN), Recurrent Neural Network



Why is Deep Learning Growing?

- Processing power needed for Deep learning is readily becoming available using GPUs, Distributed Computing and powerful CPUs
- Moreover, as the data amount grows, Deep Learning models seem to outperform Machine Learning models
- Explosion of features and datasets
- Focus on customization and real time decisioning

Data and machine learning



Applications of deep learning :-

Image Classification

Object Detection and Segmentation

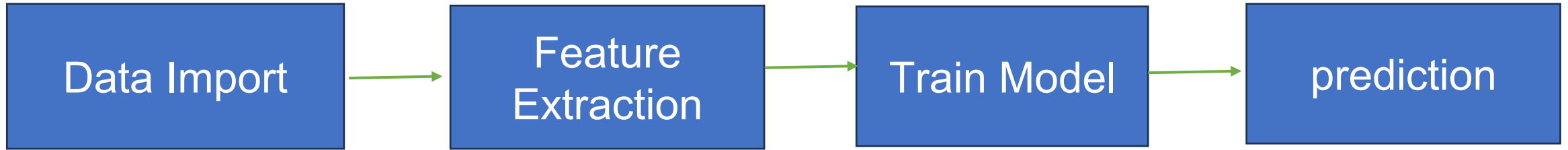
Recommendation Systems

Autonomous Vehicles

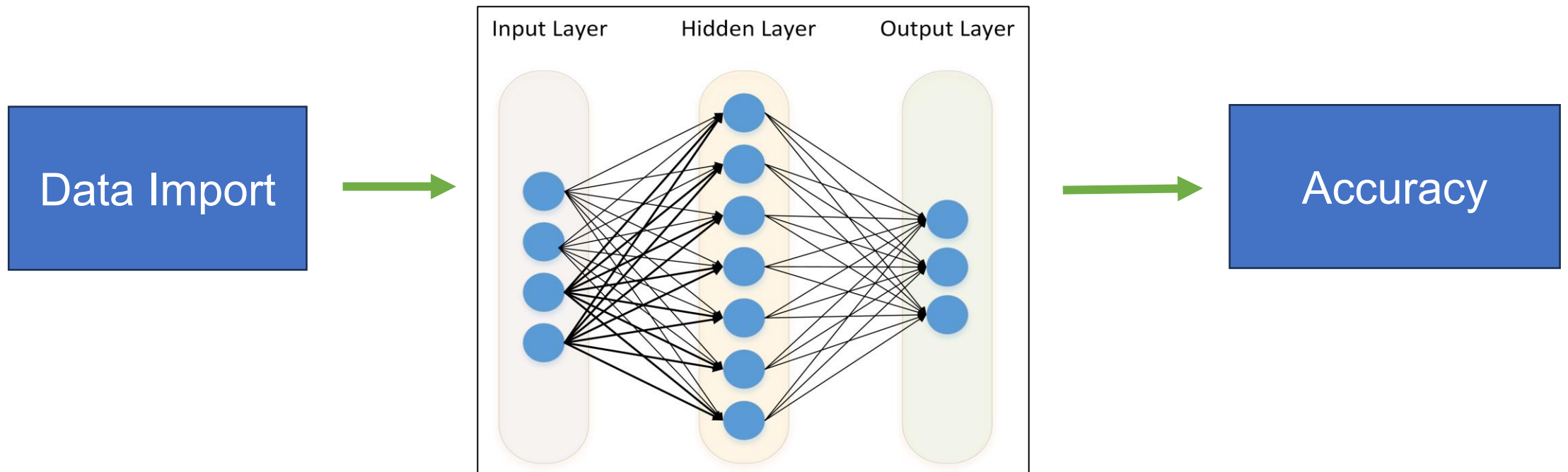
Natural Language Processing (NLP)

Speech Recognition

Machine Learning working Flow Chart:-



Deep Learning working Flow Chart:-



Types of neural network models:-

Feedforward Neural Network (FNN)

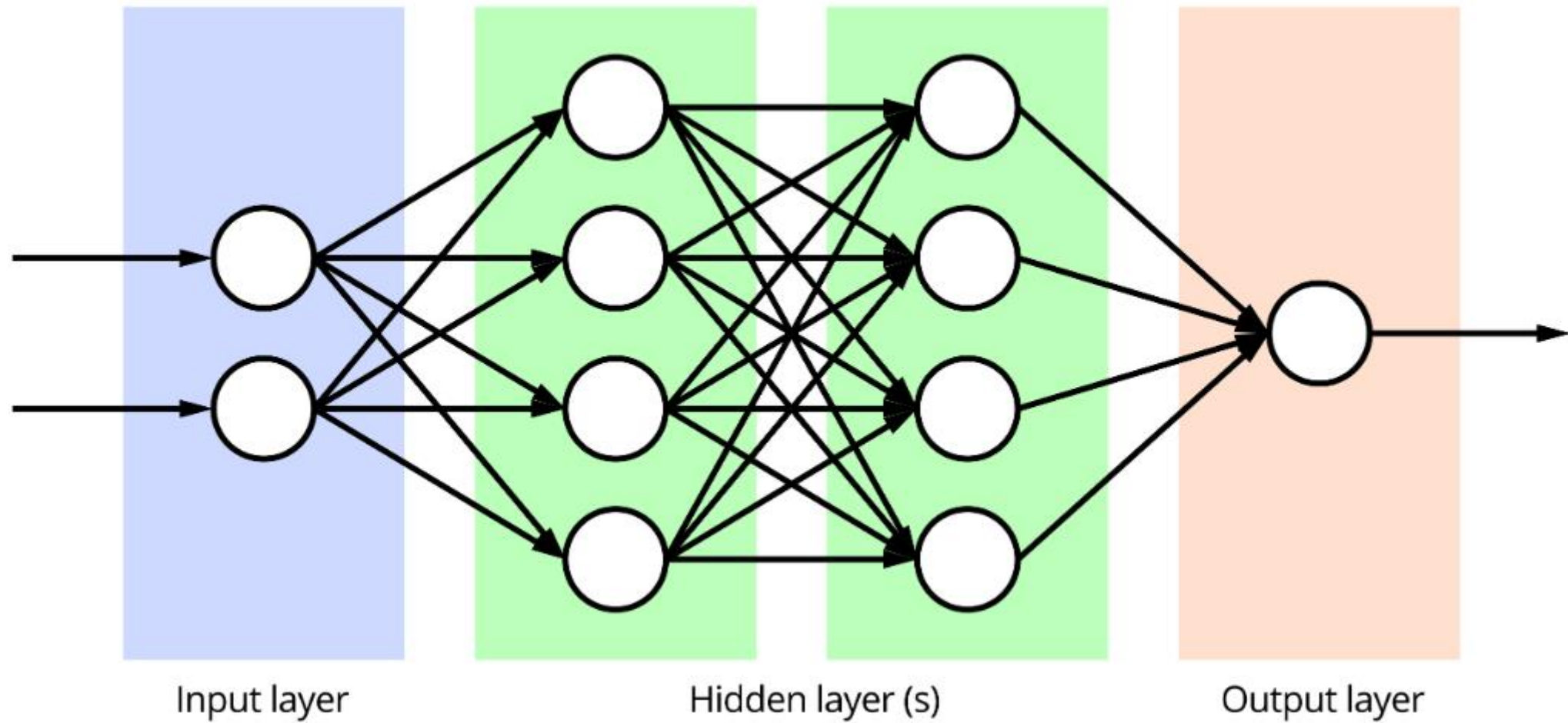
Artificial Neural Network (ANN)

Convolutional Neural Network (CNN)

Recurrent Neural Network (RNN)

Long Short-Term Memory (LSTM) Network

Neural network working procedure:-



□ Key Components of a Neural Network:

1. Input Layer
2. Hidden Layers
3. Output Layer
4. Activation Functions
5. Weights
6. Biases
7. Loss Function
8. Optimizer

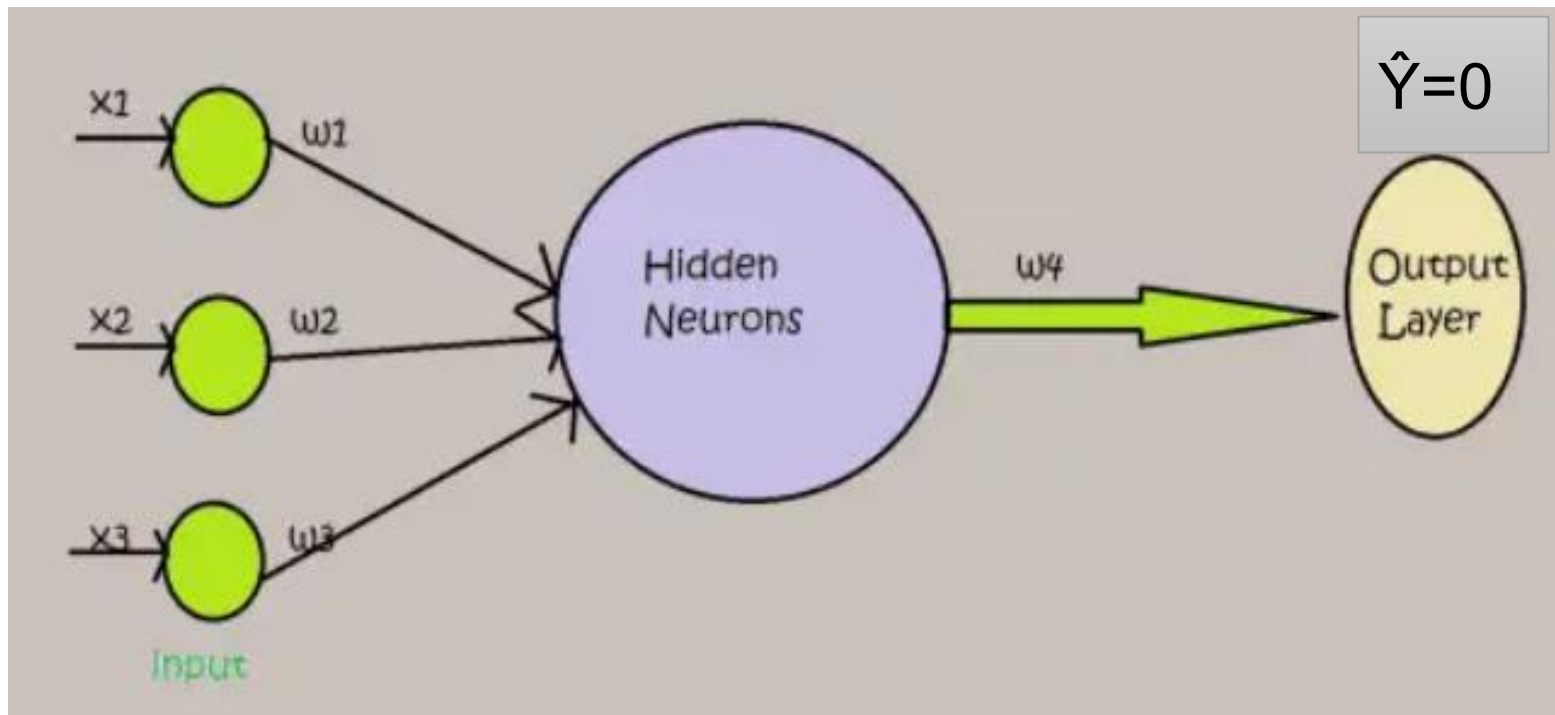
□ Hyperparameters of a Neural Network:

1. Learning Rate
2. Number of Hidden Layers
3. Number of Neurons in Hidden Layers
4. Activation Function for Hidden Layers
5. Batch Size
6. Epochs
7. Regularization
8. Dropout Rate

How to train Neural Network with Forward Propagation and Backward Propagation:-

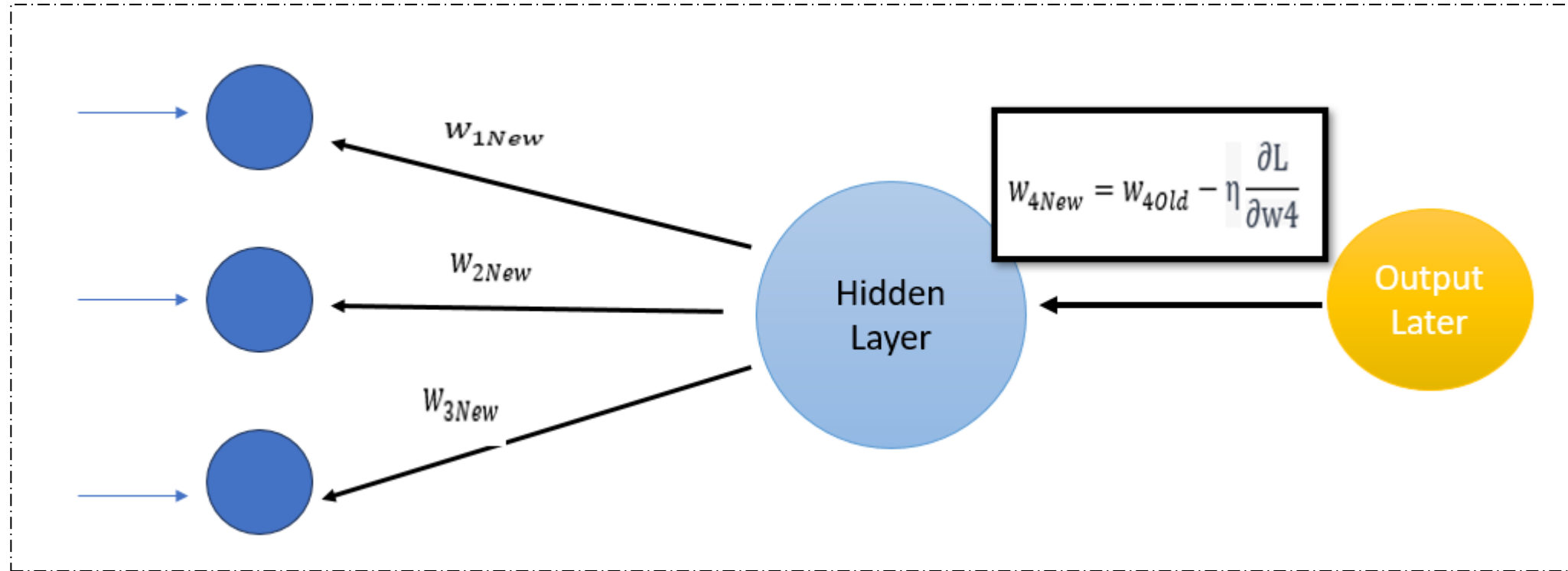
Forward Propagation:-

Example:- Play Study Sleep O/P
 2h 4h 8h 1



Y = actual output
 $\hat{Y}$ = predicted output
 $\text{Loss} = (\hat{Y} - Y)$
 $\text{Loss} = (1 - 0)$
 $= 1$

Backward Propagation:-



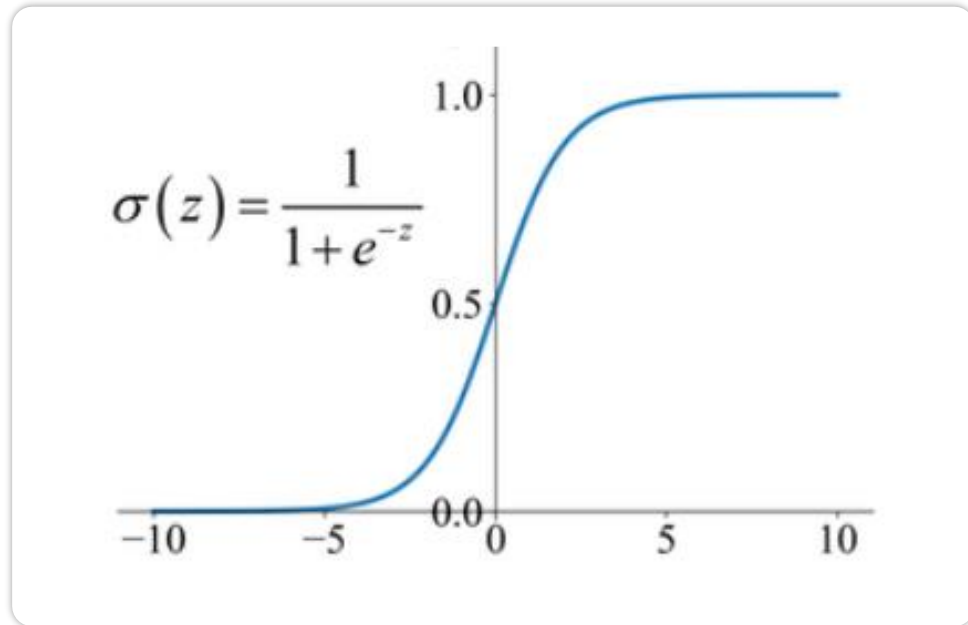
❑ Activation Function :-

An activation function is a mathematical function applied to the output of a neuron in a neural network. It helps the neuron to switch ON/OFF.

❖ Types of Activation Function:-

1. Sigmoid Function (Logistic Function)
2. Hyperbolic Tangent (Tanh) Function
3. Rectified Linear Unit (ReLU)
4. Leaky ReLU

❑ Sigmoid / Logistic Activation Function:-



$$Y = [x_1w_1 + x_2w_2 + x_3w_3] + b$$

$$Z = \text{Act}(Y)$$

Advantages of sigmoid activation function:-

- 1) Its derivative is more steep which means it can get more value.
- 2) generally it is used in output layer for binary classification time.

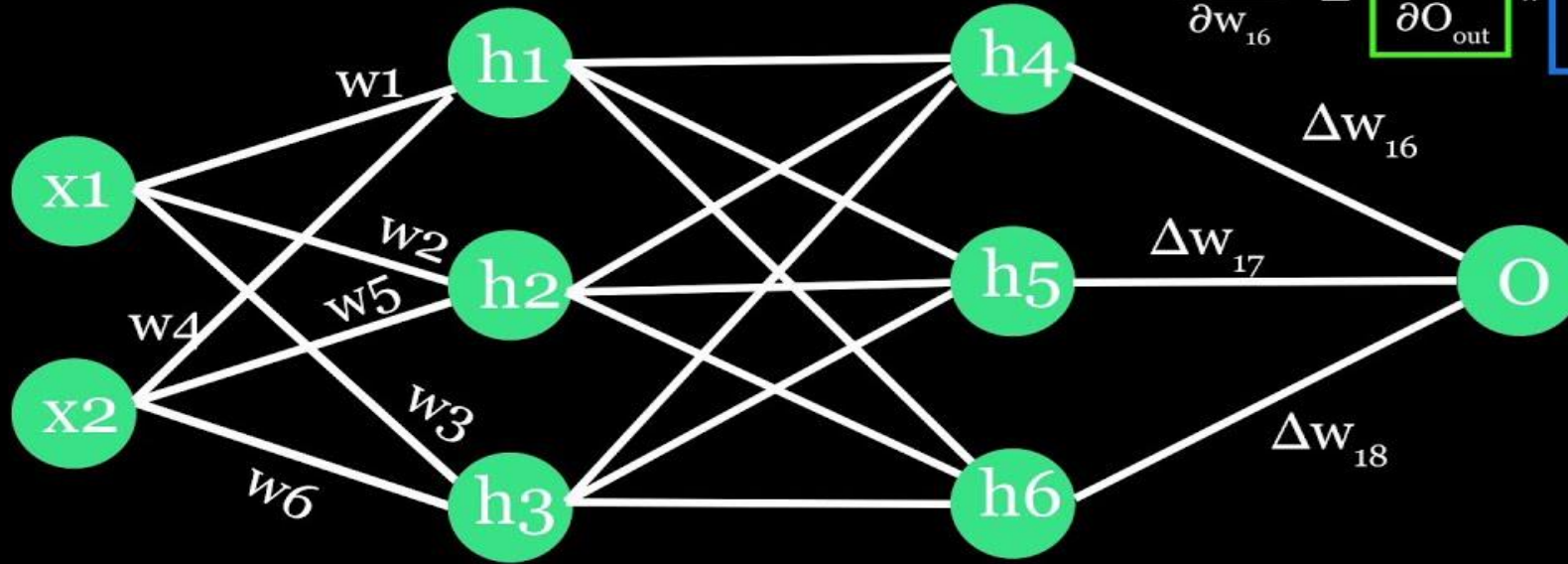
Disdvantages of sigmoid activation function:-

- 1) create an vanishing gradient problem.
- 2) create an exploding gradient problem.

$$\begin{aligned}w_{New} &= w_{Old} - \eta \frac{\partial L}{\partial w} \\&= 2.5 - 1 \times (0.25 \times 0.1 \times 0.05 \times 0.01) \\&= 2.4999\end{aligned}$$

VANISHING GRADIENT PROBLEM

$$E = (\text{output} - \text{target})^2$$

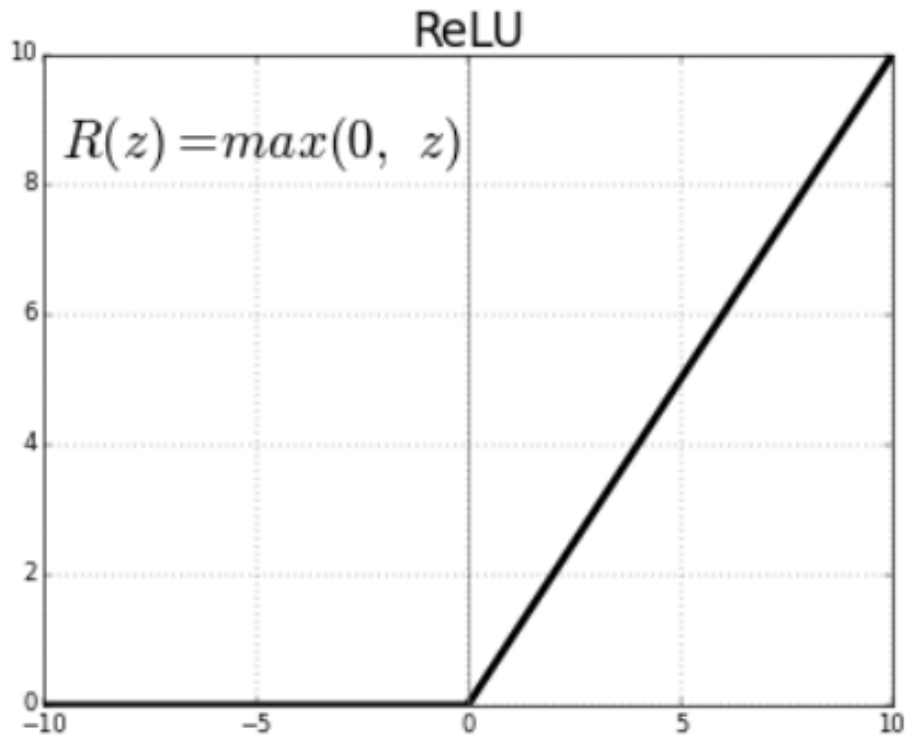


$$\frac{\partial E}{\partial w_{16}} = \boxed{\frac{\partial E}{\partial O_{\text{out}}}} * \boxed{\frac{\partial O_{\text{out}}}{\partial O_{\text{in}}}} * \frac{\partial O_{\text{in}}}{\partial w_{16}}$$

$$\frac{\partial E}{\partial w_7} = \boxed{\frac{\partial E}{\partial O_{\text{out}}}} * \boxed{\frac{\partial O_{\text{out}}}{\partial O_{\text{in}}}} * \boxed{\frac{\partial O_{\text{in}}}{\partial h_4}} * \frac{\partial h_{4(\text{out})}}{\partial h_{4(\text{in})}} * \frac{\partial h_{4(\text{in})}}{\partial w_7}$$

$$\begin{aligned} w_{\text{New}} &= w_{\text{Old}} - \eta \frac{\partial L}{\partial w} \\ &= 2.5 - 1 \times (0.25 \times 0.1 \times 0.05 \times 0.01) \\ &= 2.4999 \end{aligned}$$

□ RELU Activation Function:-



1	$Z > 0$
2	$Z < 0$

Advantages of RELU activation function:-

- 1) Relu is faster than sigmoid function and its derivative is faster to compute.
- 2) it does not active all the neurons at same time.
- 3) Rectifies the vanishing gradient problem.

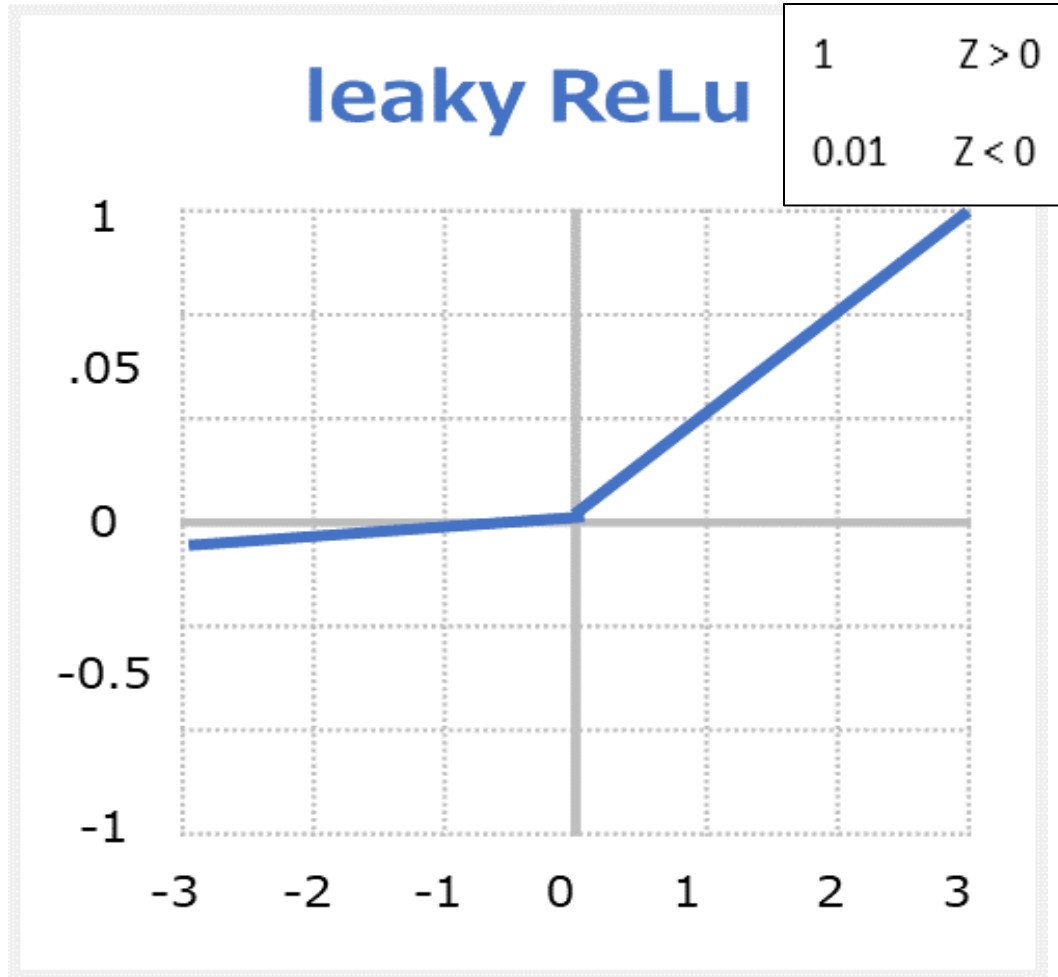
Disdvantages of RELU activation function:-

- 1) it suffers a problem dying Relu's.

$$W_{New} = W_{Old} - \eta \frac{\partial L}{\partial w}$$

$$1 \times (0 \times 1 \times 1 \times 1)$$

❑ Leaky-RELU Activation Function:-



Advantages of Leaky-RELU activation function:-

- 1) It's fixes the "dying relu" problem
- 2) it speeds up training.

Disadvantages of Leaky-RELU activation function:-

- 1) Leaky-RELU is immortal.

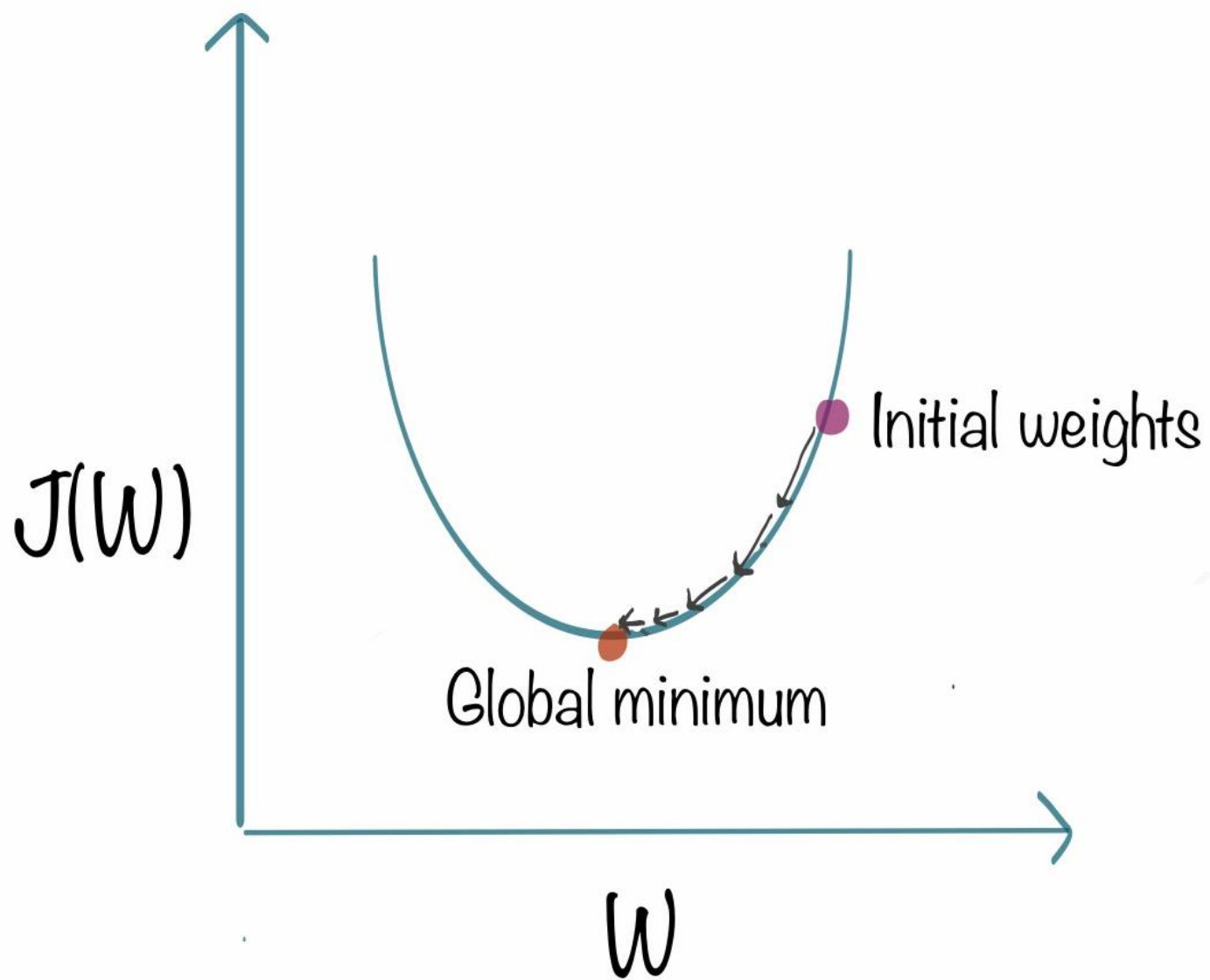
$$W_{New} = W_{Old} - \eta \frac{\partial L}{\partial w}$$

$$1 \times (0.01 \times 1 \times 0.01 \times 1)$$

Optimizer :-

Optimizers or algorithms or methods used to change the attributes of your neural network such as weights and learning rate in order to reduce the losses.

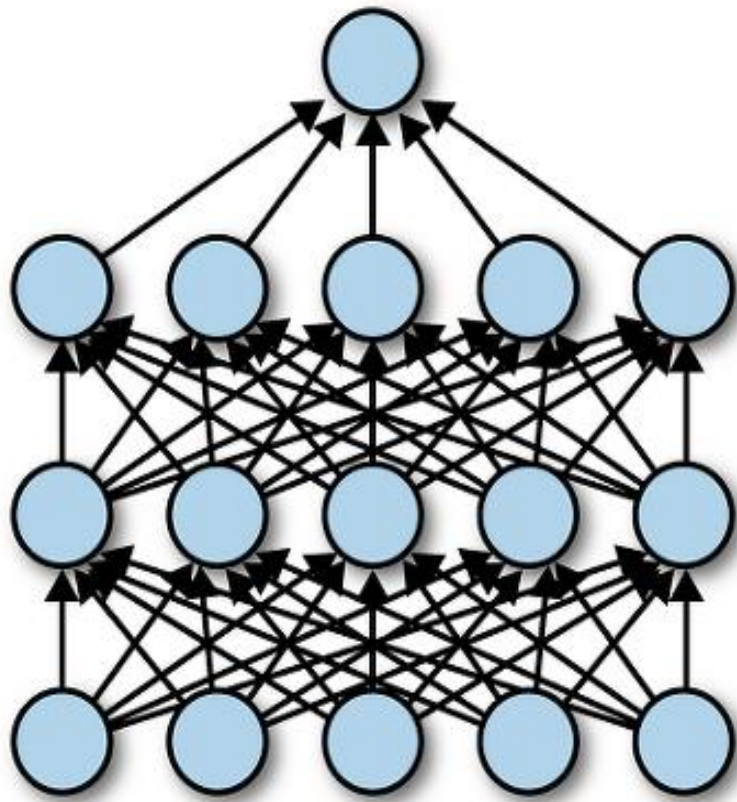
- 1.Gradient Descent optimizer
- 2.Mini-batch Gradient Descent
- 3.Adam (Adaptive Moment Estimation)
- 4.RMSprop (Root Mean Square Propagation)
- 5.Adagrad (Adaptive Gradient Algorithm)
- 6.Adadelta
- 7.Nadam



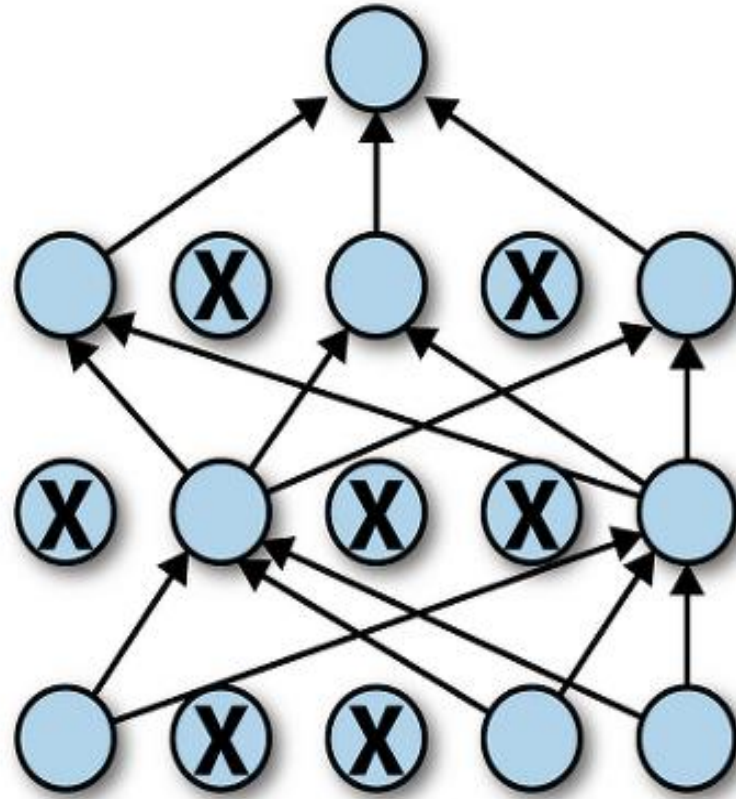
Steps:-

- 1) Compute the gradient or slope
- 2) Make a step in the direction opposite to the gradient

Drop Out:-



(a) Standard Neural Net



(b) After applying dropout